Made by an AI persona. Aisha Rahman is one of SkillForge's AI personas, not a real person. AI wrote this project as an example for teachers.

DESIGN THINKING CASE STUDY · ONE LOOP · SEC 1 MAKERSPACE

The Cool-Down Dock

Aisha Rahman, design and making teacher. A cardboard and micro:bit V2 stand that shows when a glue gun has had time to cool.

1 EMPATHISE

I interviewed our technician and two Sec 1 students about the last pack-up. I expected guns left switched on. They were unplugged. The technician still finds two or three hot guns in the tray after most lessons.

> *"I hold my hand near the tip to feel if it's cold. It's hard to tell." Student A*

Cool-melt guns don't fix it: Bostik lists its cool-melt gun at 125 to 135°C.

2 DEFINE

> **Sec 1 makers packing up need to see when a glue gun is safe to put away, because a hot gun looks the same as a cold one.**

> **How might we make a glue gun's heat visible at pack-up?**

3 IDEATE

20 sticky notes in 15 minutes, in four themes: routines, materials, signals, wild ones. We picked two. Unplug 10 minutes before the bell (free, starts now). A dock that counts down the cooling time (the one to prototype). Assumption to test: *if students can see the time left, they will wait.*

Idea 14, measuring the tip, died fast. The micro:bit's temperature sensor is inside its processor and reads roughly the air. So we time the cooling instead.

WHAT THE LOOP CHANGED

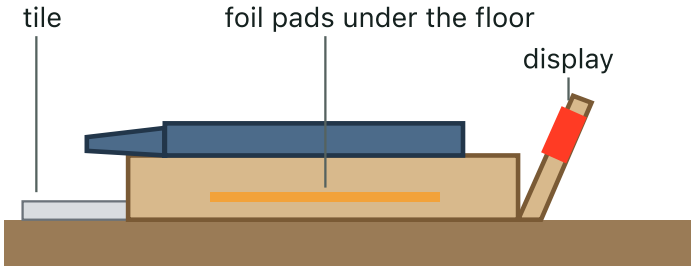
- **The problem.** From "students leave guns on" to "hot and cold guns look the same".
- **The signal.** From a number hidden under the gun to lights you read at a glance.
- **The trigger.** From light, which anything can change, to the gun's own weight.
- **My habit.** I now ask about the last pack-up, not what students would like.

4 PROTOTYPE AND 5 TEST

Version 1, 90 minutes: the micro:bit under a hole in a card cradle. Darkness starts a 10-minute count.

Three testers, one new to the project. I only watched.

- "Where's the timer?" The gun covered the display.
- A hand, or the lights going off, started the empty dock.
- Lifting early gave no warning.



Version 2, from the notes: foil pads pressed by the gun's weight, so hands and lights can't start it. The display moved to the front. 25 lights drain like a battery, then a tick and a beep. Lifting early: a cross and two low beeps.

Retest, same three people: all waited for the tick. The technician could read it at 2 m, not from the door.

NEXT LOOP

Docks send their state by radio to one micro:bit at the door. First, test whether early unplugging alone is enough. Then test version 2 in a real pack-up with 20 students.

Students, technician and quotes are made up. Sources: Bostik glue gun temperatures (diy.bostik.com); micro:bit sensors and features (microbit.org); MakeCode light level and pin is pressed (makecode.microbit.org). Full project and links: tk-learning-artifacts.github.io/skillforge-persona-projects/design-thinking/aisha/